Customer Satisfaction Score Prediction using Machine Learning

1. **Problem Statement**
   The goal of this project is to predict customer satisfaction scores based on delivery performance, pricing, and order-related features using machine learning models.

2. **Introduction**
   This project focuses on predicting customer satisfaction scores using machine learning techniques. The analysis is based on the Olist Brazilian E-commerce dataset, which contains information about customers, orders, products, and reviews.

3. **Objective**
   The objective of this project is to build a regression model that predicts customer satisfaction scores and identifies key factors affecting customer experience.

4. **Dataset Description**
   The dataset includes multiple files such as customers, orders, order reviews, order items, and products. These datasets were merged to create a unified dataset for analysis.

5. **Dataset Table**

Dataset	Description
Customers	Location details
Orders	Order timestamps
Reviews	Satisfaction score
Order Items	Price information
Products	Product category

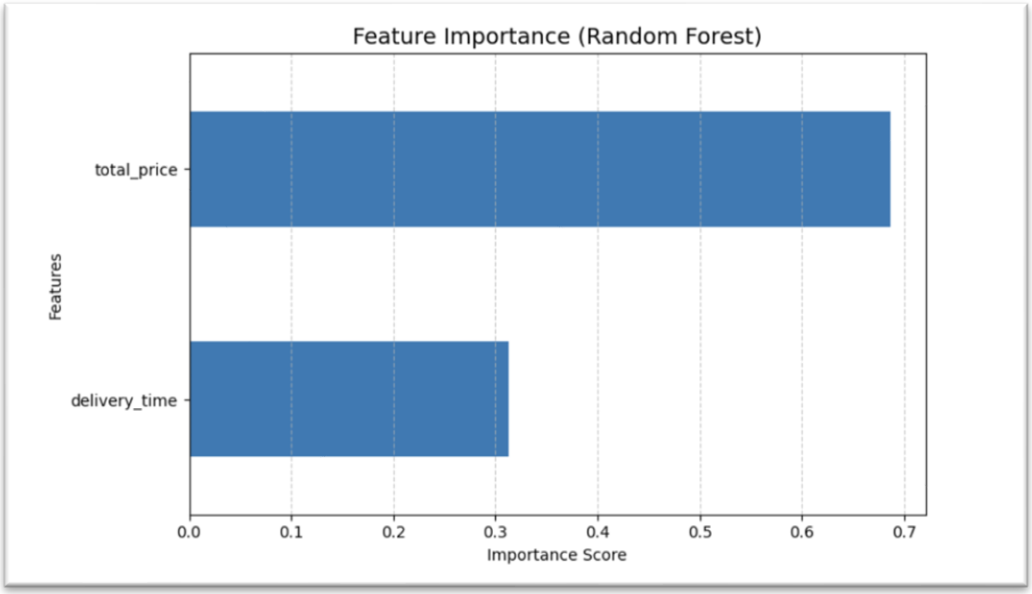
6. **Data Preprocessing**
   Data cleaning was performed by handling missing values and converting date columns into proper datetime format. Irrelevant columns were removed to improve model performance.

   - Missing values handled
   - Date columns converted
   - Irrelevant columns removed

7. **Feature Engineering**
   New features such as delivery time were created by calculating the difference between order purchase date and delivery date. Price was also used as an important feature.

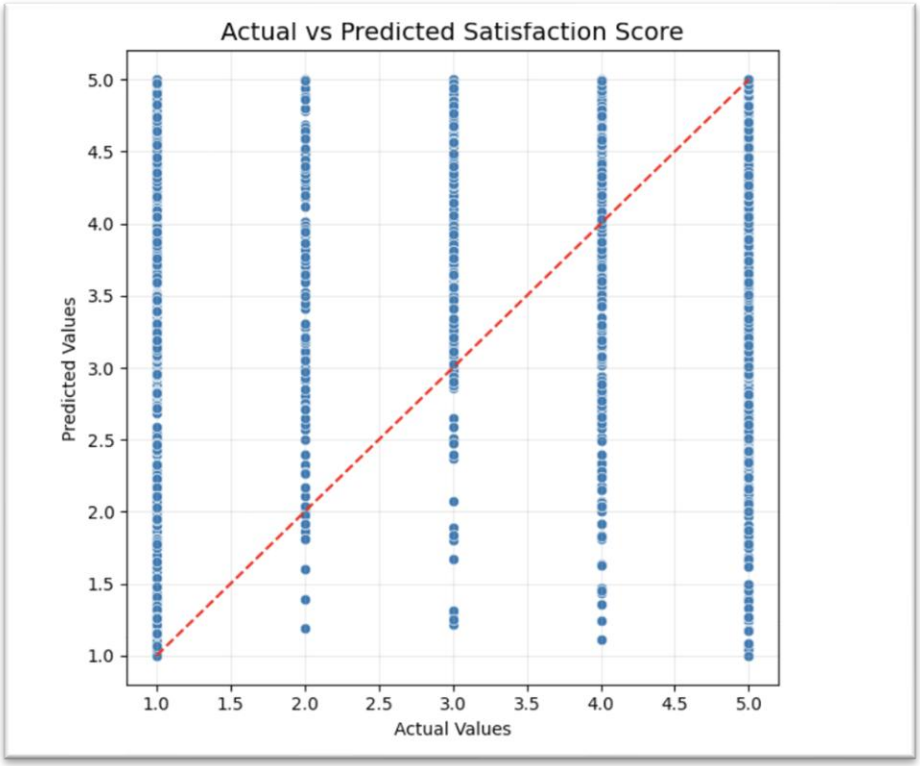
8. **Model Building**
   Three regression models were used: Linear Regression, Decision Tree Regressor, and Random Forest Regressor.

9. **Feature Importance Analysis**



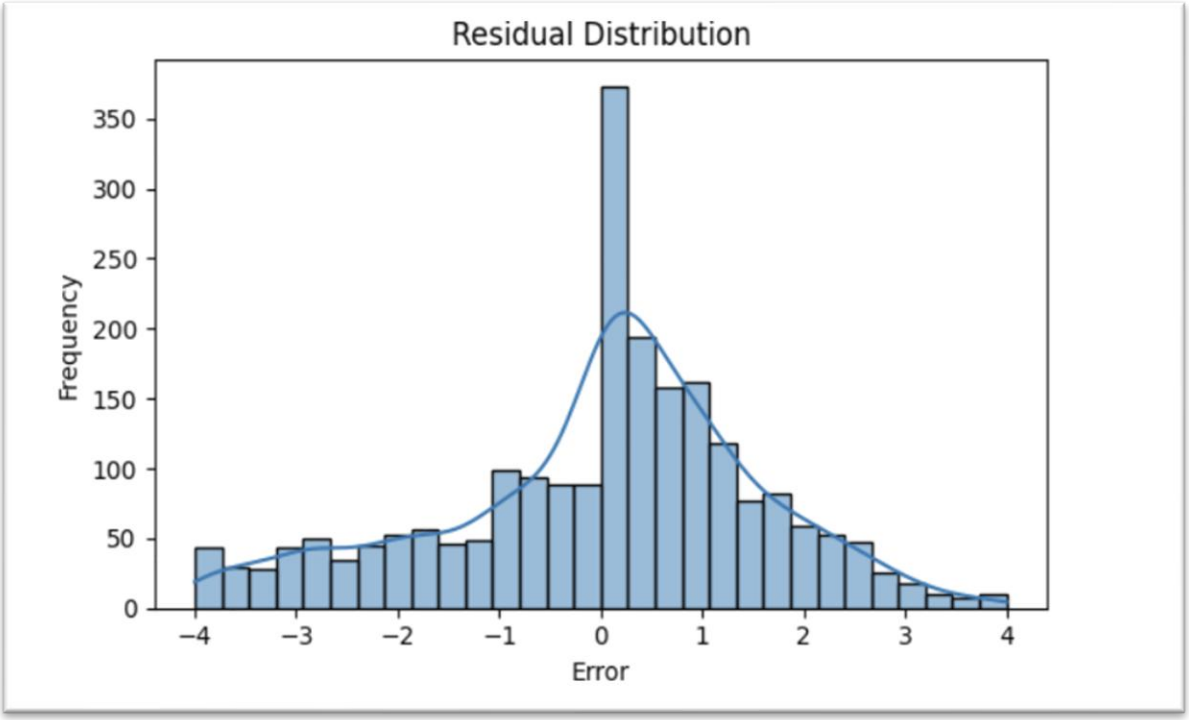
This graph shows the importance of each feature in predicting customer satisfaction.

10. **Evaluation**
Models were evaluated using Root Mean Squared Error (RMSE) and cross-validation. Random Forest showed the best performance among all models.



The graph above compares actual vs predicted values.

11. **Residual Analysis**



This graph shows the distribution of prediction errors.

12. **Key Insights**

    Delivery time has a strong impact on customer satisfaction. Faster deliveries result in higher ratings. Higher delays lead to lower satisfaction scores.

13. **Recommendations**
    Improve delivery efficiency, optimize logistics, and monitor delayed orders to enhance customer satisfaction.

14. **Conclusion**
    The project successfully demonstrates how machine learning can be used to predict customer satisfaction and provide actionable business insights.